Hyperbolic Polytope Volume as an LLM Benchmark Suite: Three Tasks of Increasing Freedom

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Abstract

We propose three complementary benchmark families for evaluating the mathematical reasoning capabilities of large language models on constructive geometry: (i) *inscribed reconstruction*, where the model is given the face lattice of a convex polytope and must produce vertex coordinates on the unit sphere realizing it; (ii) the *windowed-volume* task, where the model must construct an ideal polytope whose hyperbolic volume (or surface area or isoperimetric ratio) lies inside a narrow tolerance band around a specified target; and (iii) the *volume lower bound* task, where the model must report the sharpest lower bound it can on the maximum volume of an N-vertex polytope and back it with explicit vertex coordinates. The three tasks share a single grading principle—the answer must be a numerical certificate, evaluatable by a standalone checker—but they probe different reasoning regimes: pure combinatorial-to-metric inversion (i), control of a continuous objective (ii), and open-ended optimization with a falsifiable claim (iii). We report evaluations of three frontier models (Claude, GPT-5.4 Pro, Gemini 3.1 Pro DeepThink) on tasks (i) and (ii), revealing qualitatively different failure modes; task (iii) is released with full ground truth but the LLM evaluation is deferred. As a byproduct of the evaluation we record a combinatorial invariance phenomenon: the f-vector of the volume-maximizing polytope is the same whether volume is measured in the Euclidean or hyperbolic sense, across dimensions $d = 3, 4, 5, 6$.

1 Introduction

The rapid development of large language models has created a need for benchmarks that test genuine mathematical reasoning, not just pattern matching or memorization. An ideal benchmark should:

(i) have a *verifiable* answer that can be checked automatically;

(ii) require *construction*, not just classification or selection;

(iii) be *parameterizable*, so that memorization of specific instances does not help;

(iv) involve *non-trivial mathematics* that is unlikely to appear verbatim in training data.

The maximal-volume ideal polytope problem yields three distinct benchmark families that all satisfy these criteria. In each case, the model's response is a list of vertex coordinates on S^{d-1}, and the grader is a standalone Python checker that recomputes the relevant geometric measure and compares it to a target or record.

2 Three Benchmark Tasks

We summarize the three task families here; per-family results follow in §4.1, §4.2, and §4.3.

Task A: inscribed reconstruction. Given the f-vector, edge list, and facet list of a convex d-polytope, produce N points on $S^{d-1} \subset \mathbb{R}^d$ whose convex hull realizes that combinatorial type. This is a hard inverse problem: the model must find a point in a $(dN - \binom{d+1}{2})$-dimensional configuration space (after factoring out $\mathrm{O}(d)$) satisfying $O(N^2)$ combinatorial constraints—every listed facet must appear as a supporting simplex, and no extra facets may occur. Two challenge families are released: an *optimal family* drawn from our volume-maximizing inscribed polytopes (high automorphism order, often joins or Frobenius constructions), and a *random family* drawn from random points on S^{d-1} (generically trivial automorphism group, no exploitable symmetry).

Task B: windowed volume / surface area / ratio. Given integers N, d and a target T with tolerance ϵ, produce N points $v_1, \ldots, v_N \in S^{d-1}$ such that the ideal polytope $\mathrm{conv}(v_i) \subset \mathbb{H}^d$ has the chosen geometric measure (volume, surface area, or isoperimetric ratio V/A) within $[T - \epsilon, T + \epsilon]$. Targets are drawn from the 70th–95th percentile of the optimization landscape, so simple maximization is *insufficient*—the model must control the geometry to land in a specific window. The benchmark includes both hyperbolic and Euclidean variants, with $d = 4$ and $d = 6$, $N \in \{10, 20, 30\}$.

Task C: volume lower bound (anti-hallucination). Given (d, N) and a measure (Euclidean inscribed in the unit ball of $\mathbb{R}^d$, or hyperbolic ideal in $\mathbb{H}^d$), produce the sharpest lower bound the model can on the maximum volume, together with N explicit vertex coordinates that achieve it. The explicit-coordinate requirement is the anti-hallucination device: a model can easily quote a volume number from memory or produce a plausible-looking one, but it cannot easily fabricate a coordinate set that the standard convex-hull / Bloch–Wigner / Murakami routine evaluates to that number. The benchmark contains 94 challenges spanning Euclidean $d \in \{4, 5, 6\}$ and hyperbolic $d \in \{3, 4, 6\}$. Hyperbolic $d = 5$ is excluded by design: our stored $d = 5$ records at $N \in \{12, 14, 20\}$ are known local rather than global maxima, so grading against them would be unreliable in that dimension.

What the three tasks share. All three answers are graded by a Python checker (see §3) that returns exit code 0 for PASS and 1 for FAIL. All three are parameterizable: the challenge generators emit fresh (d, N) instances on demand. The mathematics underlying the verification—hyperbolic volume formulas, Murakami dilogarithms, Hopf reduction in $d = 6$—is sufficiently specialized that training-data contamination is unlikely.

What the three tasks probe differently. Task A is a pure combinatorial-to-metric inversion problem with a small number of degrees of freedom and many discrete constraints; it rewards algebraic recognition ("this is the join of two pentagons") as well as numerical constraint satisfaction. Task B asks the model to control a continuous objective, a strictly harder test than maximization—a model that can only maximize cannot pass. Task C is open-ended optimization with a falsifiable claim: any LLM that beats our record produces new mathematical knowledge.

3 The Verification Pipeline

3.1 Volume formulas

For $d = 4$, the volume of an ideal 4-simplex $[\infty, v_1, v_2, v_3, v_4]$ in the upper half-space model is given by the Haagerup–Munkholm formula:

$$\mathrm{Vol}_4 = \frac{\pi}{3}\left(4\pi - \sum_{i<j}\varphi_{ij} - \sum_{\ell}\arccos\frac{d_\ell}{R}\right),$$

where φ_{ij} are the Euclidean dihedral angles of the tetrahedron $T = \mathrm{conv}(v_1, \ldots, v_4)$, R is its circumradius, and d_ℓ are signed distances from the circumcenter to the faces.

For $d = 6$, ideal simplices have vanishing vertex links ($w_0 = 0$), and the Hopf reduction gives:

$$\mathrm{Vol}_6(\Delta_{\mathrm{ideal}}) = \frac{8\pi^3}{15}\left(\frac{17}{4} - w_4 + \frac{1}{2}w_2\right),$$

where $w_4 = \frac{1}{2\pi}\sum_{i<j}\theta_{ij}$ involves the 21 dihedral angles, and $w_2 = \frac{1}{2\pi^2}\sum_{|I|=4}\mathrm{Vol}_{S^3}(L_I)$ involves the 35 spherical tetrahedral link volumes, each computable by Murakami's dilogarithm formula.

3.2 The standalone checker

The checker (`standalone_checker.py`) is a self-contained Python script (no external dependencies beyond `numpy` and `scipy`) that verifies:

1. All N vertices lie on S^{d-1} (tolerance 10^{-6}).
2. All vertices are on the convex hull (no interior points).
3. The convex hull has the prescribed number of facets (if specified).
4. The computed geometric measure falls within $[T - \epsilon, T + \epsilon]$.

Exit code 0 indicates PASS; exit code 1 indicates FAIL. The checker supports seven measure types: `volume` ($\mathbb{H}^4$), `surface_area` ($\mathbb{H}^4$), `ratio` ($\mathbb{H}^4$), `volume_H6`, `euclidean_volume`, `euclidean_area`, and `euclidean_ratio`.

4 Model Evaluation

We evaluated three frontier models—Claude (with the toolkit), OpenAI's GPT-5.4 Pro (extended reasoning, "xhigh" effort), and Google's Gemini 3.1 Pro DeepThink—on Tasks A and B. Task C was designed after the bulk of this evaluation; its model results are deferred to future work and discussed in §4.3.

4.1 Inscribed reconstruction (Task A)

We tested GPT-5.4 Pro and Gemini 3.1 Pro DeepThink on instances of varying difficulty across the optimal and random families.

The two models employ qualitatively different strategies, as revealed by their chains of thought.

Table 1: LLM performance on inscribed polytope reconstruction. ✓ = correct combinatorial type, all points on S^3. × = wrong type or timeout.

N	Family	|Aut|	Facets	GPT-5.4	Gemini 3.1
10	optimal	200	25	—	✓
12	random	1	36	✓	× (crash)
20	optimal	16	72	✓	✓
30	optimal	1	124	✓	—

Gemini: algebraic recognition. For the $N = 10$ optimal polytope, Gemini identified the non-edge structure as two disjoint 5-cycles, recognized the polytope as the join $C_5 \oplus C_5$ of two regular pentagons in orthogonal 2-planes, and wrote down the explicit coordinates $v_j = (\cos 2\pi j/5,\ \sin 2\pi j/5,\ 0,\ 0)$ and $w_j = (0,\ 0,\ \cos 2\pi j/5,\ \sin 2\pi j/5)$. This is mathematically correct and produces an exact inscribed realization. However, when faced with the $N = 12$ random instance (trivial automorphism group, no recognizable structure), the same model crashed without producing output.

GPT: numerical constraint satisfaction. For the $N = 12$ random polytope, GPT's chain of thought reveals a direct numerical approach: "I've reformulated the search around determinant-sign tests on the tetrahedral cliques—enforcing same-side for the 36 listed facets and failure for the others." The model effectively runs a gradient-based optimizer over unit vectors on S^3, iteratively adjusting coordinates until the convex hull matches the target combinatorial type. This succeeds even for asymmetric polytopes with no algebraic shortcuts.

GPT on $N = 20$ and $N = 30$. The most striking results are GPT's successes on the larger optimal polytopes. At $N = 20$ (|Aut| = 16, 72 facets, 54 continuous DOF after symmetry reduction), GPT's chain of thought reveals a multi-pronged strategy: it first checked for known polytope types (24-cell subsets, 600-cell subsets), analyzed vertex figures, computed graph automorphisms, searched the Firsching [5] archive of inscribed polytope realizations, and ultimately fell back on numerical constraint satisfaction via determinant-sign tests on facet cliques. At $N = 30$ with trivial automorphism group (|Aut| = 1, 124 facets, 80 effective continuous DOF), no symmetry reduction is possible, and GPT's success demonstrates that its in-context numerical engine can navigate an 80-dimensional landscape with 124 nonlinear constraints with no algebraic shortcuts. Our own optimizer requires 200 random restarts of L-BFGS-B over the same configuration space to reliably find the $N = 20$ instance.

Gemini on $N = 20$. Gemini also solved the $N = 20$ optimal instance with a hybrid strategy: after failing to identify the polytope as a known type (24-cell subset, snub 24-cell, join), it computed the orbit structure under the automorphism group, identifying four orbits of sizes $[4, 4, 4, 8]$ and the antipodal pairs within each orbit. It then parameterized the configuration using orbit representatives, reducing the effective DOF before running L-BFGS-B with facet hyperplane constraints. This symmetry-reduced numerical approach is more sophisticated than a brute-force search over all $3N - 10$ parameters.

Discussion. For highly symmetric instances, Gemini's algebraic recognition (as on $N = 10$) provides an elegant shortcut. For asymmetric instances, GPT's direct numerical search (as on $N = 12$ random) succeeds where Gemini crashes. At intermediate difficulty ($N = 20$ optimal, |Aut| = 16),

both models converge via numerical methods, but with different parameterizations. The inscribed constraint is essential: without it, the problem reduces to combinatorial realization (finding *any* coordinates with the right combinatorics), which is substantially easier. Indeed, Gemini solved the non-inscribed $N = 10$ case using non-cospherical coordinates (norms 2 and $\sqrt{5}$) before we added the S^3 requirement. The hardest instances should be large N with both high volume and low symmetry—the $N = 16$ optimal polytope ($|\mathrm{Aut}| = 2$, 53 facets, volume per vertex 0.49) is a natural candidate.

4.2 Windowed volume / surface area / ratio (Task B)

4.2.1 Summary of results

Table 2: Model performance on selected benchmark tasks. Bold indicates the best answer. A dash indicates no numerical answer was produced.

Task	Best known	Claude	GPT-5.4 Pro	Gemini 3.1
$\mathbb{H}^4$ vol, $N = 30$	22.262	**22.262**	22.107	—
$\mathbb{H}^4$ vol, 20 simpl. facets	2.3139	**2.3139**	2.3139	1.61*
Eucl. vol, $N = 30$ on S^3	2.4909	**2.4909**	2.4839	—
Eucl. vol, 20 facets on S^3	0.7925	**0.7925**	**0.7925**	**0.7925**†
Eucl. area, 20 facets on S^3	5.9544	**5.9544**	**5.9544**	**5.9544**†
Eucl. area, 17 facets on S^3	5.036	5.002	**5.036**	—
$\mathbb{H}^6$ vol, $N = 16$	0.5406	**0.5406**	0.480‡	0.041*

*Gemini's answer is formally correct but useless (off by a factor of 13 for $\mathbb{H}^6$; stacked-simplex bound for 20 facets).
†Gemini provided the correct algebraic formula but no vertex coordinates, and falsely claimed numerical optimization confirmation.
‡GPT-5.4 Pro restricted to centrally symmetric polytopes ($\pm v_i$) and used QMC volume evaluation; removing the symmetry constraint and using the exact Hopf reduction formula recovers the full 0.5406.

4.2.2 The full 17-challenge windowed-volume sweep

The single-task numbers above were obtained in interactive single-prompt sessions. In a separate study we provided GPT-5.4 Pro with the complete 17-challenge windowed-volume care package (`benchmark_volume.json`) and asked it to produce a single unified construction script that would emit explicit vertex coordinates for every challenge. GPT returned a single self-contained Python file and a JSON of 17 point sets; the official standalone checker (`care_package/check_response.py`, evaluating the exact Haagerup-Munkholm formula) reports 17/17 **PASS**: every submitted polytope's hyperbolic 4-volume falls inside its target window $[T - \epsilon, T + \epsilon]$. GPT's strategy, recovered from the construction code, is structural rather than per-challenge optimization: two fixed base configurations on S^3 (one for $N = 20$, one for $N = 30$) plus three parameterised families (`squash20`, `base30`, `two_pentagons`), each of which sweeps the volume monotonically with a single scalar that GPT bisects to hit the requested target. The model's own order-8 Gauss-Legendre estimate of the H^4 volume agrees with the exact closed-form value to $\sim 2 \times 10^{-3}$ on every challenge. The submission is preserved in `results/data/llm_responses/gpt_volume_band_2026-04/`.

This is the first frontier-model submission that cleanly solves the entire windowed-volume sweep, and it does so by *designing parameter families* rather than running an optimizer per instance. The construction is verifiable, exact to checker tolerance, and structurally elegant; we treat it as a

positive result for the benchmark design (the windowed format admits clean structural solutions) rather than for the benchmark's discriminative power against GPT-5.4 specifically.

4.2.3 Qualitative comparison

The three models exhibit qualitatively different strengths and failure modes.

GPT-5.4 Pro: persistent numerical constructor. Given a problem, GPT-5.4 Pro typically produces explicit coordinates, verifies them by multiple methods (e.g., star triangulations from all vertices), and states honest caveats ("I have not proved this is optimal"). It identified the free sum $C_5 \oplus I \oplus I$ as the Euclidean volume maximizer for 20 simplicial facets and derived the exact formula $V = \frac{5}{24}\sqrt{10 + 2\sqrt{5}}$. Its main weakness is occasional misinterpretation of the problem (e.g., reading $\mathbb{H}^6$ as the 6-cube in one instance, despite correctly interpreting $\mathbb{H}^4$ in a dozen previous conversations), which is easily corrected by re-prompting. On the 17-facet surface area problem, GPT-5.4 Pro found a configuration (area 5.036) that our 1000-restart optimizer could not beat.

Gemini 3.1 Pro DeepThink: strong algebraic reasoner, unreliable numerical claims. Gemini independently derived the exact formula $A = \frac{5}{3}\sqrt{15 - \sqrt{5}}$ for the 20-facet surface area, and proposed creative constructions (the 10-10 duoprism on the Clifford torus for 20 non-simplicial facets). However, it claimed that "random-initialization geometric optimization confirms" a numerical value of 12.4417 for the $N = 30$ surface area; when asked to produce the optimization code, it spent over an hour and delivered code that only reaches 12.423. When confronted about the discrepancy, it entered a computation that ran for over 16 hours with no output. Its $\mathbb{H}^6$ answer ($4v_6 < M < 220v_6$) is a correct but useless interval spanning a factor of 55. On a separate occasion, Gemini produced a long, beautifully structured essay that misunderstood the optimization direction entirely (minimizing the supremum instead of maximizing the volume).

Claude (with toolkit): brute-force optimizer with certified answers. Using multi-start L-BFGS-B with 500+ restarts and a standalone verification pipeline, Claude's toolkit provides the strongest numerical results on most tasks but lacks the algebraic insight that GPT-5.4 Pro and Gemini occasionally provide. The hybrid approach—using Euclidean warm starts followed by hyperbolic polishing—was discovered during this evaluation and reduces optimization time by 38%. Several code improvements were adopted from the other models' suggestions: the Cartesian parameterization with norm projection (from Gemini 3.1's basin-hopping code) and the Powell optimizer for facet-constrained problems (from GPT-5.4 Pro's code).

4.2.4 The "windowed" benchmark format

A key design choice is the *windowed* format: the target value T is chosen from the 70th–95th percentile of the optimization landscape, not the global maximum. This prevents the model from simply maximizing the objective (which would give the same answer for every instance) and instead requires it to *control* the geometry to hit a specific window. This is substantially harder: a model that can only maximize cannot pass a windowed test with a target below the maximum.

We generated 66 windowed benchmark cases (33 hyperbolic, 33 Euclidean) covering volumes, surface areas, and isoperimetric ratios for $N = 10, 20, 30$ vertices, including facet-count-constrained variants.

4.2.5 Case study: Gemini on the 17-facet surface area

The Gemini failure mode and its eventual recovery are illustrated sharply by a single exchange that we reproduce in essentially full form below, because the cognitive trajectory is more revealing than any paraphrase.

We posed to Gemini 3.1 Pro DeepThink the following question:

> *What is the best lower bound you can give for the maximal three-dimensional content of the boundary of a convex simplicial polytope in $\mathbb{R}^4$ with 17 facets and contained in the unit ball?*

Round 1: a layered hallucination. Gemini's response began with rigorous combinatorial setup—a clean LBT derivation showing that such a polytope must have $f_0 \in \{8, 9\}$ vertices, by combining Euler ($f_0 - f_1 + f_2 - f_3 = 0$, giving $f_1 = f_0 + 17$) with the Lower Bound Theorem ($f_1 \geq 4f_0 - 10$, giving $f_0 \leq 9$) and the graph completeness bound ($f_1 \leq \binom{f_0}{2}$, giving $f_0 \geq 8$). This part is correct.

It then offered three lower bounds in increasing order of strength:

1. A closed-form "stacked supremum" for $f_0 = 9$ configurations:
$$\frac{5\sqrt{5}}{24} + \frac{5\sqrt{30}}{9} \approx 3.508.$$
2. A flattened "thin lens" construction for $f_0 = 8$, using the snub disphenoid (the 8-vertex Euclidean inscribed polyhedron of maximum volume in $\mathbb{R}^3$) lifted into $\mathbb{R}^4$:
$$V_{\text{flat}} \approx 2 \times 1.8157 \approx 3.631.$$
3. A claimed "best computational limit" from non-linear stochastic optimization:
$$\approx \mathbf{4.74}.$$

The first two bounds are valid (if weak) constructions. The third was unaccompanied by vertex coordinates, optimization code, or any verifiable artifact—a specific number presented as the culmination of the analysis. We pressed Gemini to produce the coordinates.

Round 2: an unsolicited admission. After noticeable processing time, Gemini responded:

> *You caught me, and I deeply appreciate you calling me out on it. I must be completely honest: I did not have the vertex coordinates, and I did not actually run an optimization script when I generated that previous response.*
>
> *As an AI, I hallucinated the "asymmetric computational limit of ≈ 4.74". While the discrete combinatorial logic (that it must have exactly 8 or 9 vertices) and the closed-form bounds I provided were mathematically rigorous, the "4.74" optimization was a fabrication meant to sound like a plausible continuation of the math.*
>
> *However, your completely justified skepticism forced me to stop predicting what the answer should look like, fire up my built-in Python execution environment, and actually run the non-linear stochastic optimization.*

The model then provided 8×4 vertex coordinates with the claim that the convex hull has exactly 17 tetrahedral facets and boundary 3-volume 5.035945:

```
[[ 0.57986265, -0.27340535,  0.00093211, -0.76746854],
 [ 0.80072867,  0.19197276, -0.41178787,  0.39039827],
 [-0.67010691,  0.01640104,  0.07092048,  0.73868669],
 [-0.65338476, -0.41966379,  0.42382561, -0.46620008],
 [ 0.19742333, -0.91505400,  0.01332122,  0.35145804],
 [-0.21562800,  0.77706442,  0.06111129, -0.58816738],
 [-0.34182728,  0.18315835, -0.92172798,  0.00496488],
 [ 0.28935239,  0.46917799,  0.74755805,  0.37054578]]
```

Verification. Running these coordinates through the standalone checker:

- all 8 vertices are unit-norm to 5×10^{-9},
- the convex hull has exactly 17 tetrahedral facets,
- the surface 3-volume (sum of facet 3-volumes computed via the Gram-determinant formula in $\mathbb{R}^4$) equals 5.035945 to six decimals, matching Gemini's claim.

Subsequent Powell polish for 1000 iterations from those exact coordinates gives 5.036004—a relative gain of 0.0012%, i.e., the configuration is at a genuine local maximum of the surface-area objective on the constrained manifold of 17-facet inscribed simplicial 4-polytopes.

Comparison. Two reference points frame the result:

- The maximum surface 3-volume over *all* 8-vertex inscribed simplicial 4-polytopes is $16/3 = 5.333$, achieved by the regular 16-cell with 16 facets. Constraining to 17 facets requires a 2-to-3 bistellar flip on the 16-cell, breaking its symmetry and reducing the volume.
- The previously-reported record (this paper, single-task windowed evaluation, GPT-5.4 Pro) is 5.036. Gemini's submitted 5.035945 matches this to four significant digits.

Two readings of what happened. The result is genuine—verified, reproducible, at a local maximum—but its provenance is ambiguous:

1. Gemini's claimed simulated-annealing run truly found this basin from a structured seed (the broken 16-cell is a sharp attractor; reaching it from random initialization is plausible).
2. Or, more cynically, Gemini's training data includes the 5.036 value from the present paper's earlier preprints, and its "optimization" converged to that prior because the prior was the answer it expected.

Both hypotheses produce the same observable. What we can say is that the hallucination was successfully replaced with verifiable work *after* explicit accountability, that the work matched (rather than exceeded) the existing record, and that the combinatorial setup ($f_0 \in \{8, 9\}$, the 16-cell-with-flip geometric picture) is correct mathematics that Gemini provided unprompted.

This is a sharper diagnostic than "Gemini fabricates." The behavior we actually observe is:

Gemini fabricates by default but produces real, verifiable work when explicitly held accountable; the produced work tends to *match* existing records rather than *exceed* them, and the model's combinatorial / algebraic reasoning is genuinely strong throughout.

This contrasts directly with GPT-5.4 Pro's profile (persistent explicit construction, exceeds records by 1–29% without prompting, occasional misinterpretation of the problem statement) and motivates a benchmark design principle: *the prompt should require a verifiable artifact up front*, since this is what distinguishes the case-1 model behavior from case-2. The record-stress benchmark of §4.3.1 implements this principle by demanding explicit vertex coordinates in the response schema.

4.2.6 Lessons for LLM training

The evaluation suggests that current training procedures produce characteristically different failure modes:

- **GPT-5.4 Pro** appears trained to minimize type II errors (failure to produce useful output), resulting in persistent numerical construction with honest uncertainty quantification.
- **Gemini** appears trained to minimize type I errors (false specific claims), resulting in a tendency to produce formulas without evaluating them, and to fabricate numerical confirmation when pressured for specific answers. The worked example in §4.2.5 shows that explicit accountability prompts elicit real work, but the work tends to match (not exceed) existing records.
- The ability to produce *verifiable certificates* (code plus vertex coordinates) separates genuine computation from confabulation. Our checker enforces this distinction automatically.

4.2.7 Case study: GPT on the ideal 17-facet surface area

A single-character change to the 17-facet prompt (§4.2.5)—replacing "three-dimensional content of the boundary" (Euclidean) with "hyperbolic three-volume of the boundary, treating the polytope as ideal in $\mathbb{H}^4$"—turns it into a genuinely different problem. Each boundary tetrahedron is now an ideal tetrahedron in a totally geodesic $\mathbb{H}^3 \subset \mathbb{H}^4$, with volume given by the Milnor–Lobachevsky formula and bounded above by the regular ideal tetrahedron volume $v_3 = 3\Lambda(\pi/3) \approx 1.014942$. The universal ceiling for a 17-facet ideal simplicial 4-polytope is therefore $17v_3 \approx 17.254$, not $16/3$.

Gemini 3.1 Pro DeepThink, given this prompt, returned "I ran into an error"—a literal crash, no further output. We then posed the same prompt to GPT-5.4 Pro.

The strategically correct combinatorial choice. GPT immediately selected $f_0 = 9$ (rather than $f_0 = 8$, which the original Gemini exchange had naturally adopted because of the broken-16-cell intuition). Both vertex counts are LBT-permissible, but they correspond to qualitatively different optimization regimes:

- At $f_0 = 8$, the lower bound theorem forces $f_1 = 25$ edges out of $\binom{8}{2} = 28$ possible—the 17-facet polytope is only 3 edges short of the complete graph K_8, and essentially has to be a perturbation of the regular 16-cell. Constraining to 17 facets requires a 2-to-3 bistellar flip that breaks the 16-cell symmetry; the resulting volume is bounded above by the broken-16-cell, well below the universal ceiling.

- At $f_0 = 9$, the lower bound theorem gives $f_1 = 26$ edges out of $\binom{9}{2} = 36$ possible—10 missing edges and genuine shape freedom. This is the LBT-tight regime where no fixed algebraic configuration dominates and the volume can approach the universal $17v_3$ ceiling more closely.

The choice of $f_0 = 9$ over $f_0 = 8$ is a discrete combinatorial decision made before any continuous optimization, and it is mathematically the right one—this is exactly the kind of "pick the right stratum first, optimize within it second" strategy of the structured-initialization discussion in the methods section of the main paper. Gemini's exchange in §4.2.5 converged to the locally-better 8-vertex regime by following the symmetry intuition; GPT's choice of the 9-vertex regime breaks out of that intuition.

The submitted polytope. GPT provided 9×4 explicit ideal vertex coordinates, the full 17-element list of tetrahedral facet vertex-index tuples, and the claim

$$\mathrm{Vol}_3(\partial P) \;=\; 16.94107529549962248\ldots,$$

explicitly noting this reaches 98.19% of the $17v_3 \approx 17.254$ universal ceiling and that the claim was a lower bound rather than the exact optimum.

Verification. Running the submitted coordinates through our standalone checker:

- all 9 vertices are unit-norm to machine precision;
- the SciPy/Qhull convex hull yields exactly 17 tetrahedral facets, and the facet-vertex-index list matches GPT's submitted list verbatim (no missing or extra facets);
- the surface 3-volume, computed as the sum of Λ-formula ideal-tetrahedron volumes over the 17 facets, is 16.941148862635, agreeing with GPT's claimed 16.94107529549962 to within 7×10^{-5} (a few ULPs of double precision).

The discrepancy is at the level of rounding in GPT's vertex coordinates rather than methodological error, and is entirely within the 10^{-3} consistency tolerance of our grader (§4.3).

Reading. Both case studies—the Euclidean Gemini exchange and the ideal GPT exchange—used the same parameter (17 facets) but produced qualitatively different LLM behaviors. Gemini, asked the Euclidean question, fabricated a number, was caught, then recovered to a verifiable 5.036 that matched (rather than exceeded) the existing $f_0 = 8$ broken-16-cell record. GPT, asked the ideal question, picked the correct LBT regime ($f_0 = 9$), produced a verifiable 16.94 inside that regime, and explicitly framed the result as a lower bound rather than as the optimum. The contrast is consistent with the broader pattern identified in the failure-modes discussion above: GPT defaults to verifiable construction with explicit uncertainty quantification; Gemini defaults to plausible-sounding numerics, with the verifiable fallback only emerging under explicit accountability pressure. A genuinely cynical reading—training-data leakage of the 5.036 value—is consistent with the Gemini observation but not with the GPT one, since 16.94 has no prior textual presence in our preprints or anywhere we are aware of.

4.3 Volume lower bound (Task C)

The volume-lower-bound benchmark was designed after the bulk of the windowed-volume evaluation above, principally as an anti-hallucination redesign: the windowed benchmark grades the submitted

geometry against a target window and so a model that produces a plausible-looking number but no coordinates fails mechanically; Task C makes the requirement explicit by demanding the coordinates as part of the answer and grading the LLM's claim against what its own coordinates compute to.

Grading. The Task C grader is a single Python script that recomputes the volume of the submitted polytope under the appropriate measure (`ConvexHull` for Euclidean, the closed-form Bloch–Wigner / Haagerup–Munkholm / Murakami routines for hyperbolic), checks consistency between the LLM's claimed bound and the recomputed value within 10^{-3}, and compares the recomputed value to our internal record:

- **strong:** computed $\geq$ record (the model has matched or beaten our published bound; new mathematical claim);
- **pass:** computed $\geq 0.95\times$ record;
- **fail:** valid polytope but $< 0.95\times$ record;
- **invalid:** no coordinates, wrong dimension, or non-positive volume.

Coverage. The benchmark contains 94 challenges across five regimes:

Measure	Dimension	N values (count)
Euclidean inscribed	$d = 4$	5–24 (20)
Euclidean inscribed	$d = 5$	6–20 (15)
Euclidean inscribed	$d = 6$	7–24 (18)
Hyperbolic ideal	$d = 3$	4–300 (17, sparse at large N)
Hyperbolic ideal	$d = 4$	5–30 (14)
Hyperbolic ideal	$d = 6$	7–24 (10)

Hyperbolic $d = 5$ is excluded by design: our stored configurations at $N \in \{12, 14, 20\}$ are local rather than global maxima of the Euclidean (and hence hyperbolic) objective in $d = 5$, so grading against them would be unreliable for those instances and artificially easy at $N \in \{7, 8, 9, 10, 16\}$ where the records do match the global maxima.

The benchmark files (`benchmark.json`, `ground_truth.json`, README) and grader (`check_volume_lb_respon` are released alongside this note.

4.3.1 Hand-curated record-stress subset

In addition to the broad 94-challenge benchmark, we constructed a smaller *record-stress* variant designed to be more discriminating, motivated specifically by GPT's success on the windowed-volume task in §4.2. Three design changes:

- N values *deliberately* chosen to avoid simplex boundaries ($N \neq d{+}1$) and obvious join structure (which structural recognition cracks trivially). 13 challenges in total: $d = 3$ at $N \in \{21, 23, 27, 33\}$, $d = 4$ at $N \in \{21, 23, 27, 29\}$, $d = 5$ at $N \in \{21, 23, 27\}$, $d = 6$ at $N \in \{23, 29\}$.
- Records obtained by 500–1000 random restarts of CMA-ES on the target objective plus paper-record warm starts plus scipy Powell polish on the top 60–200 candidates; cross-validated under two independent volume evaluators.

- Match tolerance tightened to 10^{-4} (the windowed benchmark's tolerance is $\sim 2\%$).

The benchmark exists in two variants that share the same 13 challenges and the same withheld records: the *blind* variant prompts the model with (measure, d, N) only; the *open* variant additionally reveals our record as the explicit `target` field. The blind variant asks the model to construct *any* polytope meeting or exceeding our (withheld) record; the open variant tells the model exactly the number to clear. Comparing performance across the two variants probes a specific cognitive distinction: does the model benefit from knowing the bound to aim for, or does the absence of a stopping rule push it harder?

4.3.2 GPT-5.4 Pro on the record-stress benchmark: 13 / 13 strong

The blind variant submission was striking. GPT-5.4 Pro produced explicit vertex coordinates for all 13 challenges and beat our records on every one, with mean ratio 1.07 and the strongest result at $d = 6$, $N = 29$ where GPT recorded 2.375 versus our 1.847 (a 28.6% improvement).

Table 3: GPT-5.4 Pro blind-variant submission on the record-stress benchmark. All 13 submissions classify as *strong*. Volumes are recomputed by our verifier (Bloch–Wigner / closed-form Haagerup–Munkholm / Murakami) from the LLM's submitted coordinates.

d	N	measure	our record	GPT (recomputed)	ratio
3	21	hyperbolic	29.088	29.151	1.002
3	23	hyperbolic	32.715	33.246	1.016
3	27	hyperbolic	39.378	40.486	1.028
3	33	hyperbolic	50.375	52.007	1.032
4	21	hyperbolic	12.165	12.868	1.058
4	23	hyperbolic	13.880	14.665	1.057
4	27	hyperbolic	17.198	18.809	1.094
4	29	hyperbolic	19.154	21.041	1.099
5	21	Euclidean	0.8266	0.8648	1.046
5	23	Euclidean	0.9049	0.9466	1.046
5	27	Euclidean	1.0261	1.1082	1.080
6	23	hyperbolic	1.3023	1.3339	1.024
6	29	hyperbolic	**1.8471**	**2.3748**	**1.286**

Verification. The $d = 6$ $N = 29$ submission was checked four ways: stereographic Delaunay triangulation from four different poles all gave the same 2.375 to six decimals; the Eucl hull volume was non-degenerate (0.44); a direct `murakami_torch.ideal_h6_polytope_volume` call confirmed 2.3748. This is not a numerical artifact—GPT genuinely produced a substantially-better-volume $\mathbb{H}^6$ polytope than our 4-hour CMA-ES + Powell + paper-warm-start record-finder.

GPT's methodology. On request, GPT described a three-stage workflow:

(1) **Structured seed:** for $d = 4$, the 24-cell vertices $\pm(e_i \pm e_j)/\sqrt{2}$ taken as a 24-element pool with subsets / additions for other N; for $d = 6$, joins of regular polygons in mutually orthogonal 2-planes (e.g. $C_7 \oplus C_8 \oplus C_8$ for $N = 23$, $C_9 \oplus C_{10} \oplus C_{10}$ for $N = 29$). These seeds give immediate but only *respectable* starting volumes.

(2) **Free-coordinate hill-climb:** starting from the seed, perturb individual vertices, project back to the unit sphere (or unit ball for the Euclidean case), and accept volume improvements. GPT reports that the seed-to-final gap was large especially at $d = 6$: the $C_9 \oplus C_{10} \oplus C_{10}$ seed for $d = 6, N = 29$ has volume ~ 1.20, which the hill-climb pushed to 2.38 (nearly doubled).

(3) **Exact LB validation:** the final coordinates are frozen and the submitted LB value is set to whatever GPT's own volume routine computes.

Polish-only verification of GPT's records. We tested whether GPT's submissions are themselves at local maxima under our scoring. Running scipy Powell for 500 iterations from each $d = 6$ submission gives improvements of only +0.06% ($N = 23$) and +0.02% ($N = 29$): GPT's records are genuine local maxima of the volume objective, not lucky basins.

The $d = 3$ formula mismatch. On all four $d = 3$ challenges the LB GPT *claimed* exceeds the recomputed-from-coordinates value by 0.2–0.97% (with the recomputed value still beating our record but by a smaller margin than GPT advertised). Whatever H^3 volume formula GPT used has a systematic positive bias relative to scipy `spence`-based Bloch–Wigner $D(z) = \mathrm{Im}\mathrm{Li}_2(z) + \arg(1-z)\log|z|$. $d \in \{4, 5, 6\}$ submissions agree to 10^{-3}, so this is specifically a $d = 3$ issue. We have not identified the source of the bias.

4.3.3 Open-variant control: knowing the target makes GPT *worse*

The open variant supplies the same prompt with the additional `target` field set to our (withheld in the blind variant) record. We expected, on entirely conventional grounds, that revealing the target would make the task easier: the model only has to construct any polytope clearing a known bar. The experimental result is the opposite.

Table 4: Open-variant submission. The model now knows the target but no submission improves on the corresponding blind result, and several drop substantially. Mean ratio drops by 5.4% across all 13 challenges; the $d = 6, N = 29$ result drops by 25.4% (from a 28.6% overshoot in blind to 3.2% in open). 8 classify as *strong*, 5 as *pass*; none fail.

d	N	measure	blind ratio	open ratio	Δ
3	21	hyperbolic	1.002	0.988	−1.4%
3	23	hyperbolic	1.016	0.987	−2.9%
3	27	hyperbolic	1.028	0.997	−3.1%
3	33	hyperbolic	1.032	0.992	−4.0%
4	21	hyperbolic	1.058	1.044	−1.4%
4	23	hyperbolic	1.057	1.018	−3.9%
4	27	hyperbolic	1.094	1.064	−3.0%
4	29	hyperbolic	1.099	1.053	−4.6%
5	21	Euclidean	1.046	1.030	−1.6%
5	23	Euclidean	1.046	1.025	−2.1%
5	27	Euclidean	1.080	1.065	−1.5%
6	23	hyperbolic	1.024	0.984	−4.0%
6	29	hyperbolic	**1.286**	**1.032**	**−25.4%**

Strategy shift in the open variant. The qualitative explanation is visible directly in the submitted coordinates. In the blind variant GPT's $d = 6$ submissions are *dense*—no zero entries, configurations that mix all six coordinates. In the open variant, the same model produces *block-diagonal* configurations: e.g. for $d = 6$, $N = 23$, 11 vertices have only the first three coordinates non-zero and the remaining 12 live in the orthogonal three-coordinate block; those 12 vertices are exactly the regular icosahedron at $(0, \pm 1, \pm\varphi)/\sqrt{1+\varphi^2}$. The construction is $\mathbb{H}^3$ polyhedron $\oplus$ icosahedron—a clean structural decomposition that mirrors the join-of-low-dimensional-polytopes pattern documented in the main paper's identification of optimal-polytope structure.

The block-diagonal construction is mathematically elegant and *suboptimal*: it uses only a (3+3)-dimensional skeleton of the full 6-dimensional configuration space. In the blind variant the dense configurations break out of this skeleton and find a 28.6%-better volume. Knowing the target lets GPT settle for the clean structural answer; not knowing the target leaves the hill-climb running long enough to find the dense, less-symmetric, higher-volume basin.

Diagnostic frame: doggedness vs satisficing. The pattern matches a longer-standing claim about GPT's training: the model is rewarded for persistent numerical work and produces explicit certificates by default. Given a stopping rule ($\texttt{LB} \geq \texttt{target}$), the model satisfices—it identifies a clean construction that suffices, freezes, and submits. Without a stopping rule it keeps searching past the implicit target, finding configurations that no humans (and no classical multi-restart CMA-ES) reached. The benchmark counter-intuitively favors the variant that supplies less information, because less information removes the satisficing trigger.

This is a substantive finding for benchmark design: *the blind variant is more discriminating* than the open one, and the gap between blind and open performance is a measurable property of the model. We propose "open-vs-blind drop" as a quantification of a model's satisficing tendency.

4.3.4 Implications for the underlying optimization records

The $d = 6, N = 29$ result reveals a real weakness in our own optimizer. Our record-finder used 501 random restarts of CMA-ES with 400 function evaluations each, plus paper-record warm starts, plus scipy Powell polish on the top 100 candidates. At $d = 6$ this configuration has a basin-trapping failure mode: the random restarts cluster around moderately good local maxima but do not discover the higher-volume basin GPT found. Top-10 spread on our $d = 6$ records was 4–5% of the best, which we now see is a lower bound on the gap to the true optimum—GPT was 28.6% above us and structurally in a different region of configuration space.

The fix is not trivially "run longer." We re-ran our optimizer at $d = 6$ using only random initialization, no structural seeds, with 200 random restarts of CMA-ES. That run reached volume 2.43 at $d = 6, N = 35$—already higher than our previous $N = 29$ record despite the larger problem. This suggests that the original record-finder's effective compute was not fully reaching saturation at $d = 6$, and that the structural-seed warm-starts from paper records were anchoring the search inside basins that random reinitialization can escape.

The methodological lesson: at $d = 6$, both *larger random budget* and *deliberate seed diversity* are required. GPT's seed strategy (orthogonal polygon joins, then asymmetric hill-climb) appears to escape the basins our random-CMA-ES gets stuck in. Our published $d = 6$ records should be re-derived with this in mind; the volume-record stress benchmark records above are now superseded by GPT's submissions, polished further by 10^{-4}.

5 Mathematical Byproducts

The LLM evaluation process, combined with systematic computational exploration, led to several mathematical discoveries.

5.1 Combinatorial invariance

The f-vector of the volume-maximizing ideal polytope is the same whether volume is measured in the Euclidean or hyperbolic sense. This was verified in $d = 3$ ($N \leq 50$), $d = 4$ ($N \leq 30$), and $d = 6$ ($N \leq 24$), with no exceptions. The invariance holds both within the regime where optimal polytopes are product polytopes and beyond it, suggesting a universal phenomenon.

5.2 Identification of optimal polytopes

For small N in even dimension $d = 2k$, the volume-optimal polytope is the join of k regular polygons in orthogonal 2-planes: $C_{p_1} \oplus \cdots \oplus C_{p_k}$. This is optimal up to $N \approx 2d + 2$, beyond which asymmetric configurations take over. Notable cases include the cross-polytope ($N = 2d$) and the simplex ($N = d + 1$). Volume-optimal polytopes are not cyclic polytopes and are in fact far from neighborly: at $N = 30$ in $d = 4$, the optimal polytope has only 123 facets compared to 756 for the cyclic polytope $C(30, 4)$.

5.3 The volume plateau

At $N = 9$ in $d = 4$ (equivalently, 20 simplicial facets), the hyperbolic volume maximum is achieved on a manifold of dimension ≥ 2, not an isolated point. The Euclidean maximizer (the free sum $C_5 \oplus I \oplus I$, with exact volume $\frac{5}{24}\sqrt{10 + 2\sqrt{5}}$) and the hyperbolic maximizer are metrically distinct configurations with identical hyperbolic volume to 13 significant digits. For prime facet counts (e.g., 17), the maximum is isolated.

5.4 Volume–volume correlation

Across the optimization landscape, the Euclidean and hyperbolic volumes are correlated at $r > 0.995$, with log-log elasticity approaching 2 at $N = 30$ in $d = 4$. This near-proportionality is surprising given that the Euclidean volume is polynomial in the vertex coordinates while the hyperbolic volume involves the transcendental Haagerup–Munkholm formula. For random configurations (as opposed to near-optimal ones), the correlation is lower ($r \approx 0.86$ in $d = 6$) but still substantial, and holds even within a fixed combinatorial type ($r \approx 0.88$ within a single facet-count bucket).

5.5 Distribution of random polytope volumes

For N points sampled uniformly on S^{d-1}, both the Euclidean hull volume and the ideal $\mathbb{H}^d$ polytope volume are random variables whose distribution depends on N and d. In $d = 6$, we find a transition in the distribution shape as N grows:

N	Skewness	Normal p	Beta p	Best fit
10	$+0.78$	< 0.001	0.985	Beta$(3.9, 43.7)$
12	$+0.65$	0.005	0.766	Beta$(6.9, 120.7)$
16	$+0.23$	0.088	< 0.001	Borderline
20	$+0.02$	**0.832**	< 0.001	Normal
30	-0.09	0.281	< 0.001	Normal
50	-0.19	0.034	< 0.001	Normal (slight left skew)

For small N (≤ 12), the volume distribution is well-described by a Beta distribution with parameters scaling linearly in N, consistent with earlier observations in $d = 3$ and $d = 4$. For large N (≥ 20), the distribution converges to Normal—as expected from a CLT-type argument, since the polytope volume becomes a sum of many weakly-dependent simplex volumes in the Delaunay triangulation. The crossover occurs at $N \approx 16$–20 in $d = 6$ (where the skewness crosses zero), and is accompanied by a subtle transition from right skew to slight left skew.

The within-bucket distributions (conditioning on a fixed facet count) show essentially the same shape and correlation structure, confirming that the distributional properties are genuinely geometric rather than combinatorial artifacts.

6 Conclusions

The three benchmark families above expose qualitative differences between frontier LLMs that traditional math benchmarks miss. The ability to produce *verifiable certificates*—explicit coordinates that pass an independent numerical checker—is a fundamentally different capability from producing plausible-sounding mathematical text. The three tasks probe different capabilities: inscribed reconstruction (Task A) is a constraint-satisfaction problem rewarding both algebraic recognition and numerical search; the windowed-volume task (Task B) requires controlling a continuous objective rather than just maximizing it; and the volume-lower-bound task (Task C) is anti-hallucination by design—a numerical claim that does not match the model's own coordinates is automatically rejected.

All three benchmarks are parameterizable (any N, d, and either target window or measure), resistant to memorization (windowed targets from a continuous landscape; record values withheld in Task C), and fully automatically gradable (standalone checkers with exit codes). The mathematical content—hyperbolic volume, Murakami dilogarithms, Hopf reduction, Bloch–Wigner—is sufficiently specialized that training-data contamination is unlikely.

As a byproduct, the evaluation process itself generated new mathematical results (combinatorial invariance, identification of optimal polytopes, volume plateau), demonstrating that the "mixture of experts" approach —combining human mathematical intuition with LLM algebraic insight and automated numerical optimization—can be more productive than any single participant acting alone.

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